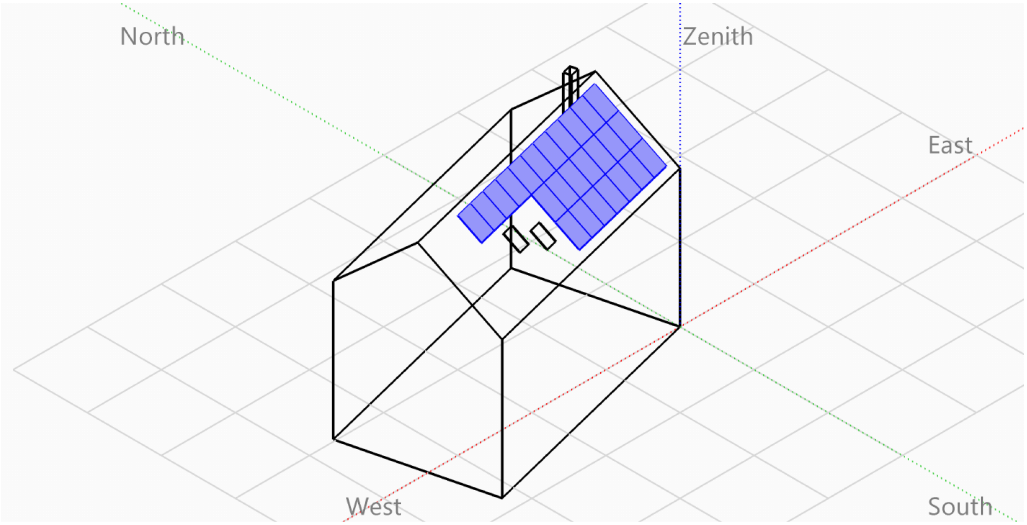
10.9 kWp rooftop PV with battery for a high-demand family house

From a real roof to a financed decision: load analysis, hand sizing of array, strings, inverter and battery, a 3D shading scene generated in AutoCAD, hourly PVsyst simulation of two cases (without and with storage) and a 25-year economic model in Excel.

Site — single-family house in the Nürnberg region, Bavaria

Load — high-demand household, 9,000 kWh/year

Tools — Hand calculation · AutoCAD + AutoLISP · PVsyst 8.1 · Excel



3D near-shading scene — house, chimney, skylights and 25 modules (PVsyst)

10.88 kWp	10.0 kW	9.84 kWh	1,155	39 → 65 %	7.3 yr
25 × SOLARWATT 435 Wp	SMA STP 10.0 Smart Energy	SMA Home Storage	kWh/kWp specific yield	autarky without → with battery	payback with battery

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Concept study on a real building; depth scaled to fit the array. Not for construction.

Same site and orientation, a deeper roof for 25 modules

Location & climate

Location	Nürnberg region, Bavaria
Altitude	≈ 340 m
Weather data	Meteonorm 9.0
Global horizontal (GHI)	1,154 kWh/m²
Mean ambient temp.	9.4 °C
Design T_{min} / T_{max}	−21.4 / 35.8 °C
Grid	400 V, 3-phase

Roof dimensions from public map data; exact location withheld for privacy. Irradiation: PVsyst / Meteonorm.

Building

Type	detached house (scaled)
Footprint	16.48 × 9.40 m
Ridge height	13.19 m
Roof	gable, tilt 40°
Ridge axis	76° / 256°
PV roof face azimuth	−14° (SSE)
Slope length eave → ridge	6.14 m

Real depth 8.14 m; the model uses 9.40 m so the 25 modules fit (page 05). Eave height 9.25 m. Obstacles: chimney behind the ridge, two roof windows on the south face.

Design basis

Household	high-demand
Annual consumption	9,000 kWh
System type	on-grid, hybrid
Grid code	VDE-AR-N 4105
Single-phase limit	4.6 kVA
Tariff	EEG, 2 tiers (> 10 kWp)

Three values are **assumptions**: roof tilt 40° (not surveyed), consumption 9,000 kWh (twice the 4,500 kWh reference case, e.g. a two-generation household or electric hot water with home office) and a house depth of 9.40 m instead of 8.14 m, so that the array the load needs fits on one roof face.

Size by hand first, then let the simulation check it

01

Site & load data

Roof dimensions and orientation from public maps; consumption set to twice the reference household.

BayernAtlas · Google Earth · Stromspiegel 2025

Output: design basis, pages 02–04

02

Hand sizing

Array size from daily energy, string window from temperature-corrected voltages, inverter ratio, battery from night demand.

Datasheets · calculator

Output: 25 modules, STP 10.0, 9.84 kWh

03

3D model

An AutoLISP script draws house, obstacles and modules from the measured dimensions and writes a Collada scene.

AutoCAD · AutoLISP

Output: shading scene (.dae)

04

PVsyst simulation

Hourly simulation with near shading, custom module and battery files, two variants: without and with storage.

PVsyst 8.1 · Meteonorm 9.0

Output: yield, losses, energy flows

05

Economics

25-year cash flow per case from the simulated energy flows; payback, NPV, IRR, LCOE and sensitivity.

Excel model (formula-driven)

Output: investment decision

Two cases, same PV array

Component	Without battery	With battery
PV array	25 × 435 Wp, 2 strings (10 + 15)	same
Inverter	STP 10.0 Smart Energy	same
Storage	none	Home Storage 9.8 (9.84 kWh)
Investment	13,050 €	18,462 €

The hybrid inverter is used in both cases, so the battery can be retrofitted later. The comparison isolates the value of storage alone.

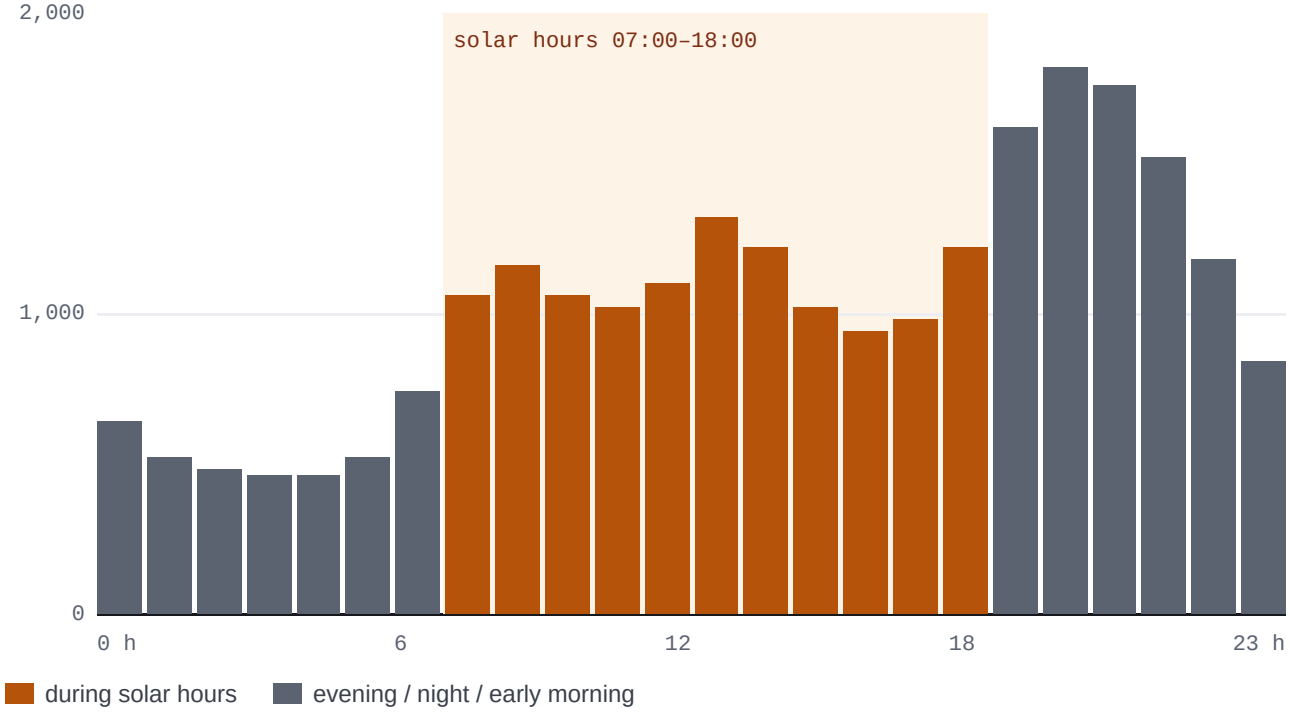
Questions the study answers

1. How many modules does a 9,000 kWh household need, and how large must the roof be to hold them?
2. How should 25 modules be split over an inverter with two unequal MPP inputs, from −21 °C to +67 °C cell temperature?
3. How large should the battery be, and is it compatible with the inverter?
4. How much energy does the system produce, and how much of it does the house actually use?
5. Does the battery pay for itself?

Half of the daily demand falls outside solar hours

Hourly load, typical day (W)

24.66 kWh/day · constant over the year



How the profile was built

High-demand scenario: twice the 4,500 kWh reference household (Stromspiegel 2025, 4 persons). Every hour of the H0-shaped profile is multiplied by 2, so the shape and the 51 % night share stay the same; only the energy doubles.

Annual demand	9,000 kWh
Daily demand = 9,000 ÷ 365	24.66 kWh
Average power	1,028 W
Peak hour (19 h)	1,820 W
Energy 07:00–18:00	12.10 kWh (49 %)
Energy outside 07:00–18:00	**12.56 kWh (51 %)**

The 51 % night share is the number that sizes the battery (page 08). Without storage, PV can never cover it; this is why self-consumption stays low without a battery.

Limitation: one profile for all seasons. A real 9,000 kWh house (heat pump, EV) would be more winter- or evening-heavy than a scaled H0 profile.

25 modules — and a roof deep enough to hold them

Energy-based sizing

E_{day} = 9,000 kWh ÷ 365 = 24.66 kWh

E_{req} = 24.66 × 1.3 (losses) = 32.05 kWh

P_{PV} = 32.05 kWh ÷ 3.0 h (PSH) = 10.68 kWp

N = 10,685 W ÷ 435 W = 24.56 → **25 modules**

P_{DC} = 25 × 435 Wp = **10.875 kWp**

Peak sun hours (PSH) of 3.0 h/day and a 1.3 loss factor are conservative design values. Rounded up, never down: the array must at least meet demand.

Cross-check with yield data

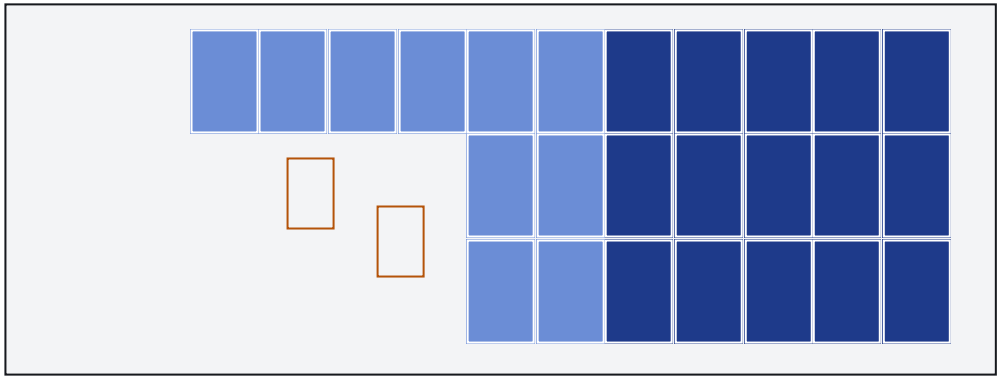
Source	kWh/kWp	Annual kWh	vs demand
Design rule (1 kWp → 1,000 kWh)	1,000	10,875	121 %
Global Solar Atlas, optimum tilt	1,126	12,245	136 %
PVsyst, this roof	**1,155**	**12,559**	**140 %**

Oversizing is intended: winter output is low, and the summer surplus charges the battery.

Module: SOLARWATT Panel classic AM 2.5 (435 Wp)

Pmax	Efficiency	Vmp / Imp	Voc / Isc
435 Wp	**22.3 %**	**32.6 V / 13.4 A**	**39.0 V / 13.9 A**
β Voc	**γ Pmax**	**NMOT**	**Size**
−0.25 %/K	**−0.31 %/K**	**45 °C**	**1722 × 1134 mm**

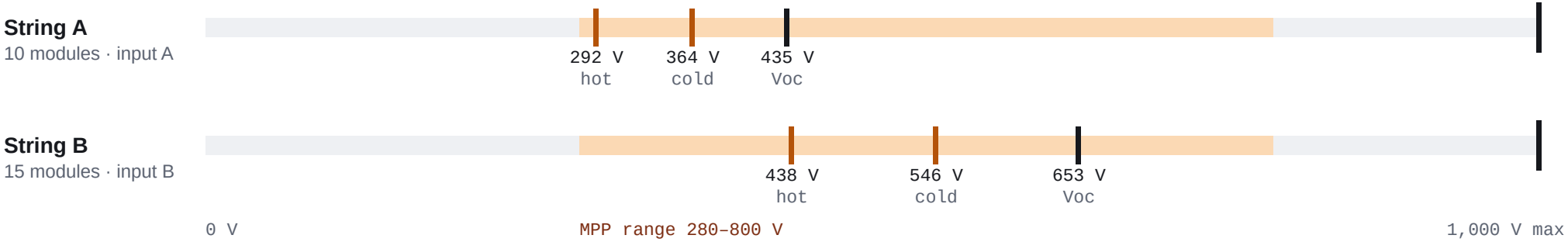
Roof fit, south face seen from the garden



W gable — 16.48 m · ridge at top, eave at bottom — E gable

Portrait, 7 columns × 3 rows + 4 along the ridge row; 0.5 m edge and 0.4 m eave/ridge clearance; roof windows (orange) kept free. ■ string B (15) ■ string A (10). The real 8.14 m deep house (slope 5.31 m) fits only 2 portrait rows; at 9.40 m the slope is 6.14 m and 3 rows fit.

Two strings, 10 + 15, matched to the unequal MPP inputs



Window per module (as 5 kW case)

$V_{oc,cold}$ (−21.4 °C) = 43.52 V

$V_{mp,cold}$ = 36.38 V

$V_{mp,hot}$ (T_{cell} 67.1 °C) = 29.17 V

N_{max} = 1,000 / 43.52 = 22.98 → **22**

N_{min} = **280** / 29.17 = 9.6 → **10**

The STP 10.0 starts its MPP range at 280 V (210 V on the STP 5.0), so the shortest string is now 10 modules.

Which split of 25?

13 + 12	5.66 kWp on input A (PVsyst share 3.3 kWac): overload ≈ 9 %
9 + 16	9 × 29.17 = 263 V < 280 V on hot days
10 + 15	**1 : 2, same ratio as the inputs**

Checks per input

Condition	A (10)	B (15)	Limit A / B
$V_{oc,cold}$	435 V	653 V	< 1,000 V
$V_{mp,hot}$	292 V	438 V	> 280 V
$V_{mp,cold}$	364 V	546 V	< 800 V
DC power	4.35 kW	6.53 kW	6 / 12 kW
I_{sc}	13.9 A	13.9 A	20 / 40 A
I_{mp}	13.4 A	13.4 A	12.5 / 25 A

Input A: 292 V on hot days keeps a 4 % margin above 280 V. I_{mp} is slightly above 12.5 A, as in the 5 kW case; I_{sc} is within 20 A, so this only means minor clipping on cold, clear days. Input B has headroom in every row.

PVsyst result: overload loss on input A **0.07 %** of the annual yield.

THE STRINGS

A — input A, 10 modules

B — input B, 15 modules

	A	B
Power, kWp	4.35	6.53
V_{mpp} STC	326	489
V_{mpp} range	292–364	438–546
V_{oc} −21 °C	435	653
Range allowed	**10–22**	**10–22**

A 10 kW three-phase hybrid inverter at DC/AC 1.09

AC power from the array

$P_{AC} = P_{DC} \div (1.1 \ldots 1.3)$

$10{,}875 \div 1.3 = 8{,}365\ W \ldots 10{,}875 \div 1.1 = 9{,}886\ W$

→ **10.0 kW**, DC/AC = 10,875 ÷ 10,000 = **1.09**

On-grid, the inverter is sized from the array, not from the load: the grid covers peaks and surges. A slightly smaller inverter is optimal because the array rarely reaches its STC rating.

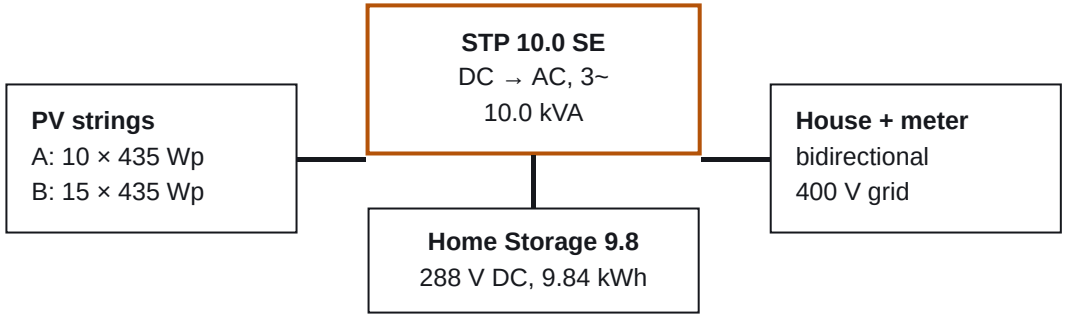
Options considered

Inverter	DC/AC	Verdict
SMA Sunny Boy (1-phase)	–	no: 1-phase limit 4.6 kVA
SMA STP 8.0 Smart Energy	1.36	no: ratio above 1.3
2 × STP 5.0 Smart Energy	1.09	no: two units, two batteries
SMA STP 10.0 Smart Energy	**1.09**	**selected**

SMA STP10.0-3SE-40, key data

Rated AC power **10,000 VA**	Max. PV power **15,000 Wp**	Max. DC voltage **1,000 V**
MPP range **280–800 V**	Input current A / B **12.5 A / 25 A**	Isc max A / B **20 A / 40 A**
Battery voltage **150–600 V**	Battery current **30 A (U > 220 V)**	Usable DC power A / B **6 kW / 12 kW**

System single-line



The two MPP inputs are unequal (6 kW / 12 kW usable DC power). PVsyst models this as 3.3 + 6.7 kWac; the string split on page 06 follows the same 1 : 2 ratio. Simulated clipping: 0.07 %. Night consumption: 1.55 % of the yield.

9.84 kWh: the rules of thumb set the size, not the full night

1. Upper bound: night demand

E_{night} = 24.66 × 50 % = 12.3 kWh

$E_{PV,day}$ = 10,875 × 1,000 ÷ 365 = 29.8 kWh

Surplus = 29.8 − 12.3 = 17.5 ≥ 12.3 ✓

C_{max} = 12.3 ÷ η 0.945 = **13.0 kWh**

2. Lower bound: rules of thumb

1 kWh per MWh/yr → **9.0 kWh**

≤ 1 kWh per kWp → 10.9 kWh

Target range → **9.0 – 13.0 kWh**

3. Modules of 3.28 kWh

9.0 ÷ 3.28 = 2.74 → **3 modules = 9.84 kWh**

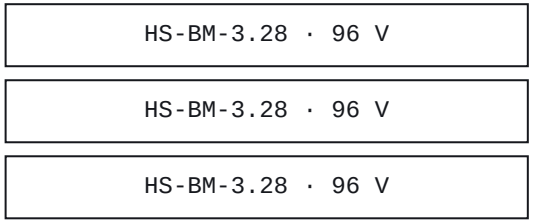
4 modules (13.1 kWh) would match the full night but exceed both rules; the 4th module would mainly cycle on summer nights. Chosen: the smallest size inside the range, expandable later.

Compatibility checks

Check	Value	Result
Approved for STP-SE	HS 6.5–16.4	OK
Empty: 3 × 90 V ≥ 150 V	270 V	OK
Full: 3 × 108 V ≤ 600 V	324 V	OK
U_{bat} > 220 V for full 30 A	≥ 270 V	OK
Current = min(36, 30)	30 A	OK
Power = 288 V × 30 A	8.64 kW	OK
C-rate = 8.64 ÷ 9.84	0.88 C	OK
Usable vs average night	9.15 / 12.3 kWh	74 %

At 3 modules the pack voltage is always above 220 V, so the inverter's full 30 A is available, which was only a note in the 2-module case. 8.64 kW is far above the 1.82 kW peak hour; this is the value entered in PVsyst for charging and discharging.

SMA HOME STORAGE 9.8

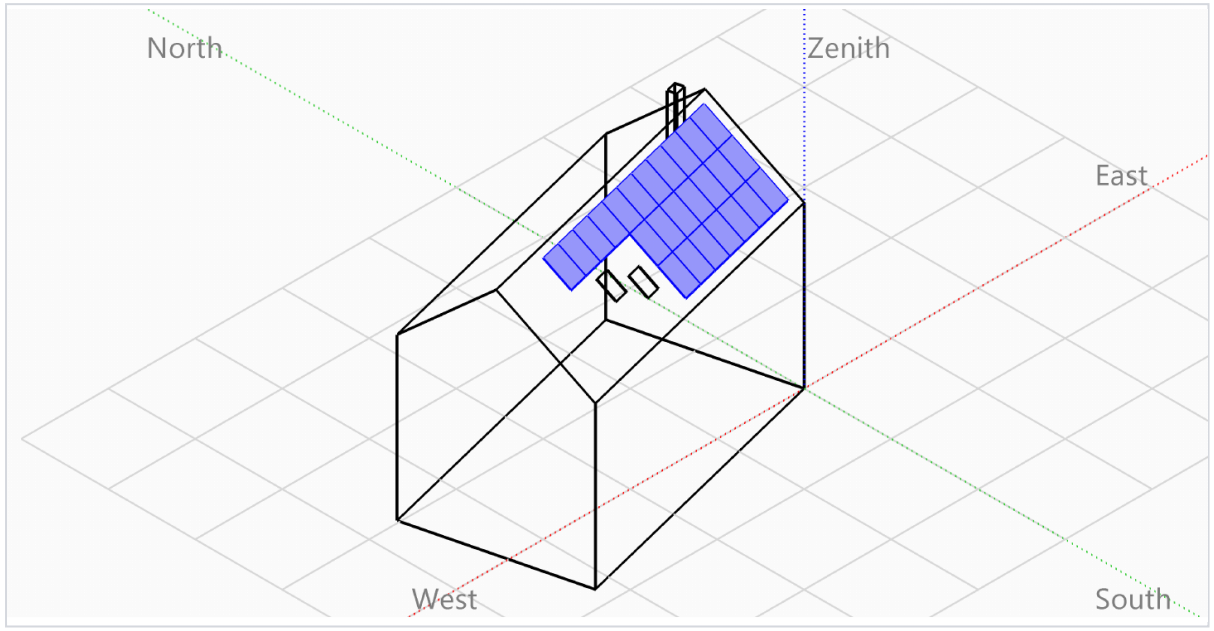


3 modules in series → 288 V, 34.17 Ah

Nominal energy	9.84 kWh
Chemistry	LFP, 90 cells
Max. current	36 A
Efficiency	94.5 %
Cycle life	8,000
Warranty	10 years
Expandable to	**16.4 kWh**

Custom PVsyst battery file (Battery block, 30 cells, 96 V, cut-off 108 / 90 V). SOC window 5–98 % → 9.15 kWh used.

Same script, deeper house: near-shading loss of 0.16 %



PVsyst Construction/Perspective after import — 25 PV fields, 48.8 m², tilt 40° / azimuth −14°

−0.16 %	−2.79 %	+16.2 %	−0.6 %
near shading	IAM (glass)	tilt gain vs GHI	vs optimal orientation

Re-using the 4,500 kWh script

1. **Parameters** — only three values changed: house depth 9.40 m, N = 25, portrait orientation. Chimney position is now set relative to the ridge.
2. **Geometry** — house extruded from the gable profile; eave height derived: 13.19 − 4.70 × tan 40° = 9.25 m.
3. **Module placement** — modules drawn flat, then transformed onto the 40° plane 10 cm above the tiles; obstacles skipped with 0.2 m clearance.
4. **Orientation** — model rotated 104° so the PV face points to azimuth −14°.
5. **Export** — a Collada (.dae) file written directly by the script, PV top surface as its own material, imported into PVsyst without Blender or MeshLab.

Debugging step: PVsyst first placed both sub-arrays on the main MPPT input and counted 0.7 inverters. Assigning the 15-module sub-array to the secondary input gave 1 inverter, 10 kWac, DC/AC 1.09.

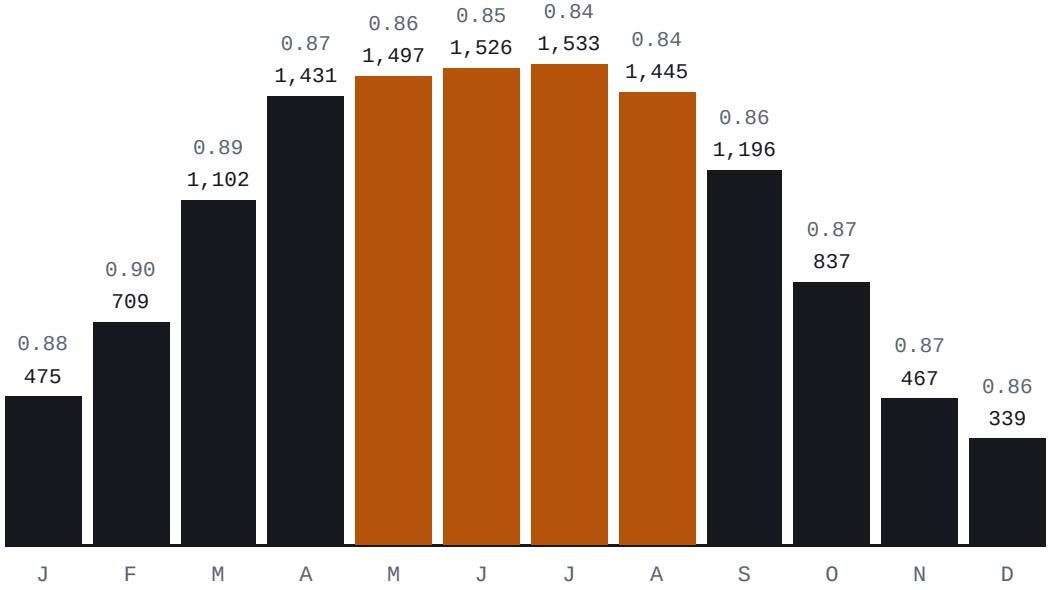
Why the loss is low: the chimney sits north of the ridge and the roof windows are east of the array, so neither casts a shadow on it while the sun is in front of the plane.

12,559 kWh per year at a performance ratio of 86 %

12,559 kWh	1,155	86.1 %	1,341	11,887 kWh
AC energy, year 1	kWh/kWp specific yield	performance ratio	kWh/m² on the plane	P90 (variability 4.2 %)

Monthly AC energy

kWh · PR above each bar



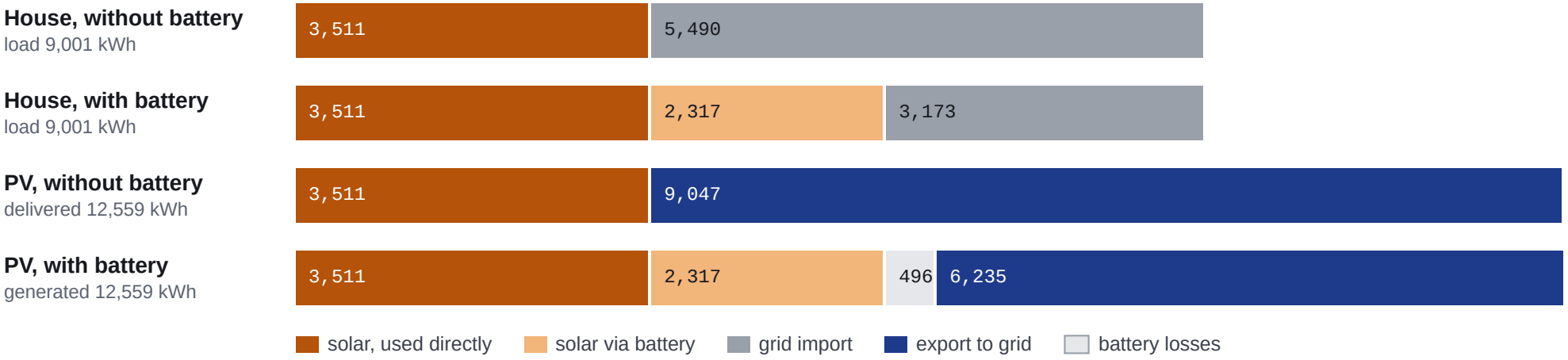
July delivers 4.5 × the December yield. PR is higher than in the 4,500 kWh case (86.1 vs 84.4 %): the inverter's fixed night draw is spread over twice the energy (1.55 vs 3.00 %).

Where the energy goes (loss diagram)

Global horizontal irradiation	**1,154 kWh/m²**
Transposition to 40° / −14°	+16.2 %
Near shading	−0.16 %
IAM (angle of incidence)	−2.79 %
Array nominal energy at STC	**14,154 kWh**
Low irradiance	−1.11 %
Module temperature	−3.51 %
Module quality (+5 W tolerance)	+1.25 %
Mismatch + DC wiring	−2.94 %
Inverter efficiency	−3.83 %
Inverter clipping (DC/AC 1.09)	−0.07 %
Inverter night consumption	−1.55 %
AC energy delivered	**12,559 kWh**

Thermal model: semi-integrated, U_c = 20 W/m²K (modules 10 cm above tiles). Night consumption of the hybrid inverter = 1.55 %.

The battery moves 2,300 kWh a year from the grid to the sun



Key indicators (PVsyst, year 1)

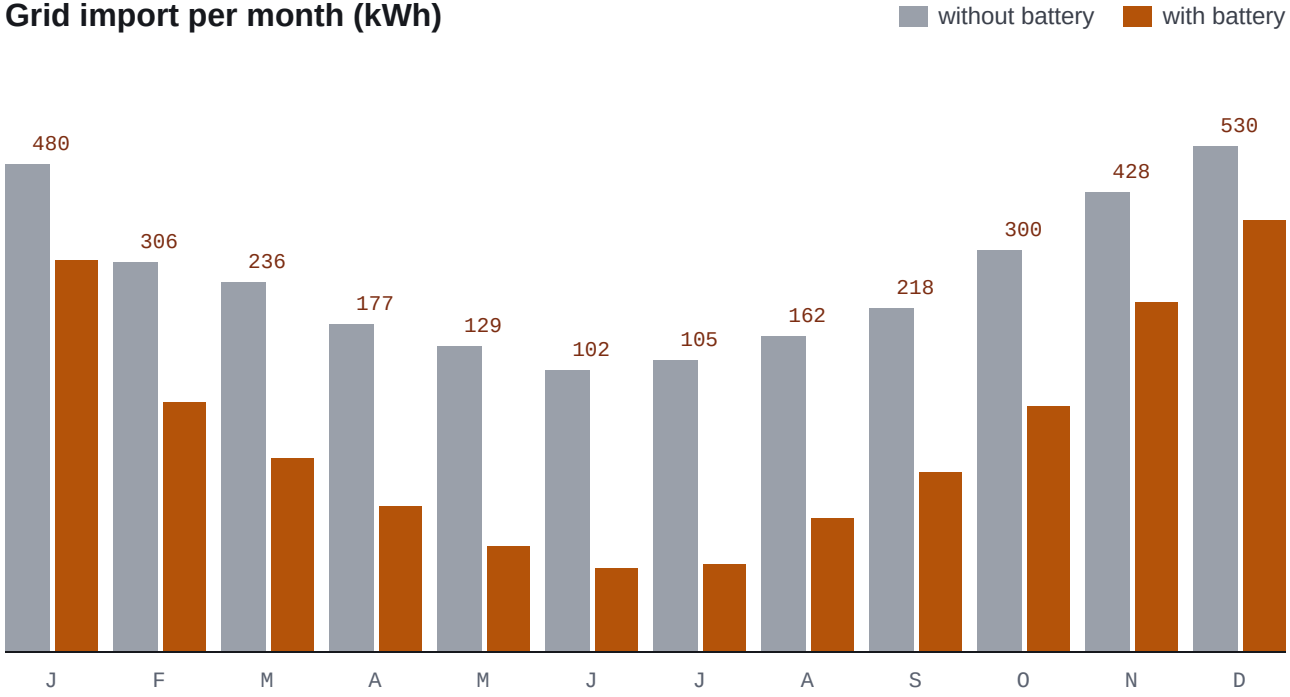
kWh / year	Without battery	With battery	Change
Solar used in the house	3,511	5,828	+2,317
Grid import	5,490	3,173	−2,317
Export	9,047	6,235	−2,812
Autarky = solar used ÷ load	39.0 %	**64.7 %**	+26 pts
Self-consumption = used ÷ delivered	28.0 %	48.3 %	+20 pts

Reading the balance

1. Midday surplus that went to the grid is now stored: export falls by **2,812 kWh**.
2. The battery returns **2,317 kWh** in the evening and night, replacing grid purchases one to one.
3. The difference, **≈ 495 kWh**, is lost in charging, storage and discharging (≈ 82 % round trip for the full DC→AC path).
4. About 2,317 ÷ 9.15 ≈ **253 equivalent full cycles** per year, far below SMA's 8,000-cycle rating.

The night outlasts the battery, even in summer

Grid import per month (kWh)



Monthly autarky with battery

J	F	M	A	M	J	J	A	S	O	N	D
37%	56%	69%	76%	83%	86%	86%	79%	71%	61%	42%	31%

What the months show

Month	PV kWh	Import w/o	Import with
July	1,533	358	**105**
April	1,431	401	177
October	837	492	300
December	339	620	**530**

From May to August the battery is full every evening but empty before morning: about 3.4 kWh per night still comes from the grid, because the average night (12.6 kWh) is larger than the 9.15 kWh the battery can deliver. Summer autarky peaks at 86 % (95 % in the 4,500 kWh case). In December the array produces 339 kWh against a 764 kWh load, so autarky only rises from 19 % to 31 %.

Consequence for design: a 4th battery module (13.1 kWh) would close part of the summer night gap but not winter. It is worth a separate PVsyst variant before deciding (page 17).

The hand design held; the energy estimate was 15 % conservative

Quantity	Hand calculation	PVsyst	Deviation	Why
Specific yield	1,000 kWh/kWp	1,155 kWh/kWp	+15.5 %	design rule ignores the 16 % tilt gain
Peak sun hours (after losses)	3.0 h/day	3.16 h/day	+5 %	1,155 ÷ 365
Annual AC energy	10,875 kWh	12,559 kWh	+15.5 %	follows from yield
Performance ratio	0.80 (assumed)	0.861	+0.06	low shading, small night draw
DC/AC ratio	1.09	1.09	0	
V_{oc} at −21 °C, A / B	435 / 653 V	437 / 655 V	+0.4 %	PVsyst model β_{oc} −0.27 %/K
Vmpp at 50 °C, A / B	306 / 458 V	299 / 449 V	−2.2 %	PVsyst uses γ Vmp −0.32 %/K
Clipping, input A	2.0 % (sizing)	0.07 %		sizing estimate is a worst case
Battery energy to house	9.15 kWh × 365 = 3,340	2,317 kWh	−31 %	little surplus in winter months
Autarky with battery	≈ 76 % (ideal day)	64.7 %	−11 pts	0.49 × 0.8 + 9.15 ÷ 24.66, no winter

Electrical design: confirmed

Voltages agree within 2.2 %, DC/AC is identical, and clipping on the small input is 0.07 % instead of the 2 % sizing estimate. No change needed.

Yield: hand rule is conservative

The 1,000 kWh/kWp rule underestimates this roof by 15.5 %; Global Solar Atlas (1,126) and PVsyst (1,155) agree within 3 %.

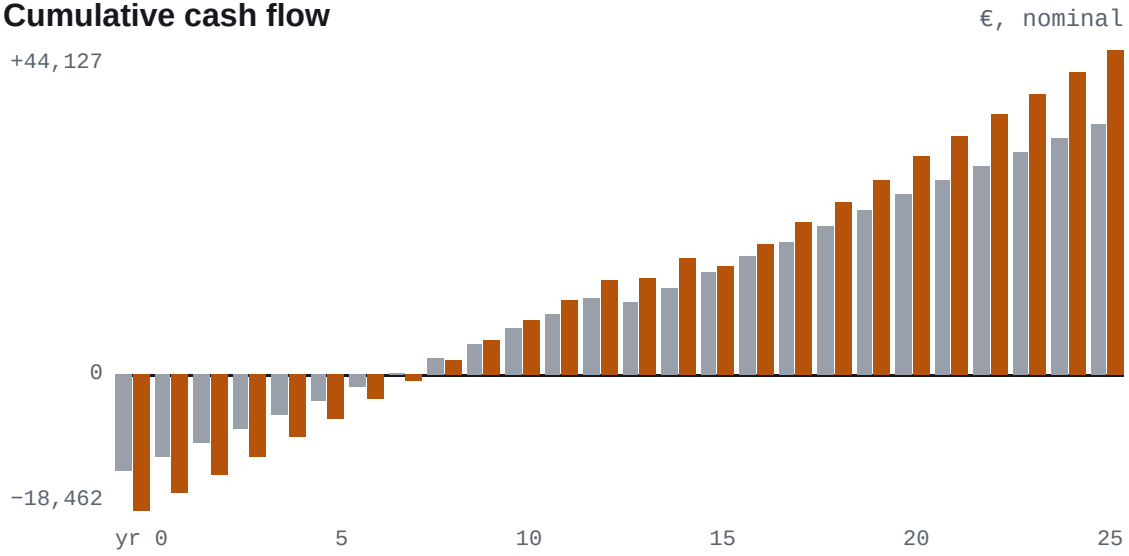
Battery: limited by winter

The hand method assumes one full cycle every day. The simulation delivers 69 % of that: full cycles in summer, small ones in winter when there is little surplus.

Both cases pay back in 7 years; the battery adds 5,100 € of value

25 years, nominal €	Investment	Net cash, yr 1	Payback	NPV at 3 %	IRR	LCOE	Cumulative yr 25
Without battery	13,050	1,859	6.9 yr	+19,651	13.8 %	8.5 ct	+34,066
With battery	18,462	2,447	7.3 yr	**+24,752**	12.7 %	13.4 ct	**+44,127**
Battery alone (difference)	5,412	588	8.5 yr	+5,101	9.9 %	–	+10,060

Cumulative cash flow



Dips in year 13 (inverter replacement, both cases) and year 15 (battery replacement). After year 20 the EEG tariff ends and export earns only the 4 ct market value.

Assumptions (net prices, 2026)

PV system, turnkey	1,200 €/kWp	size effect vs 1,300
Battery, installed	550 €/kWh	42watt 07/2026
Grid price	37 ct, +2 %/yr	BDEW 08/2026
Feed-in (EEG, blended)	7.63 ct, 20 yr	7.71 / 6.67 ct tiers
Export after year 20	4.0 ct	assumption
O&M incl. insurance	1 %/yr, +2 %	assumption
Inverter replacement, yr 13	2,500 €	assumption
Battery replacement, yr 15	400 €/kWh	assumption
Degradation	0.5 %/yr	assumption
Discount rate / lifetime	3 % / 25 yr	household
Energy flows	PVsyst yr 1	pages 10–11

Every kWh used at home saves 37 ct; every kWh exported earns 7.6 ct. The battery's value is the spread between the two, on about 2,300 kWh a year.

Year by year with battery: the investment is back in year 8

n	Year	Self-used kWh	Export kWh	Savings €	Export €	Costs €	Net €	Cumulative €
0	2026	–	–	–	–	–	**−18,462**	−18,462
1	2027	5,828	6,235	2,156	476	185	**2,447**	−16,015
2	2028	5,799	6,204	2,188	473	188	**2,474**	−13,541
3	2029	5,770	6,173	2,221	471	192	**2,500**	−11,041
4	2030	5,741	6,142	2,254	469	196	**2,527**	−8,514
5	2031	5,712	6,111	2,288	466	200	**2,554**	−5,960
6	2032	5,684	6,081	2,322	464	204	**2,582**	−3,378
7	2033	5,655	6,050	2,356	462	208	**2,610**	−768
8	2034	5,627	6,020	2,392	459	212	**2,639**	1,871
9	2035	5,599	5,990	2,427	457	216	**2,668**	4,539
10	2036	5,571	5,960	2,463	455	221	**2,697**	7,237
11	2037	5,543	5,930	2,500	452	225	**2,728**	9,964
12	2038	5,515	5,901	2,537	450	230	**2,758**	12,722
13	2039	5,488	5,871	2,575	448	2,734	**289**	13,011
14	2040	5,460	5,842	2,614	446	239	**2,820**	15,831
15	2041	5,433	5,812	2,652	443	4,180	**−1,084**	14,748
16	2042	5,406	5,783	2,692	441	248	**2,885**	17,633
17	2043	5,379	5,754	2,732	439	253	**2,918**	20,550
18	2044	5,352	5,726	2,773	437	259	**2,951**	23,501
19	2045	5,325	5,697	2,814	435	264	**2,985**	26,486
20	2046	5,299	5,669	2,856	433	269	**3,020**	29,506
21	2047	5,272	5,640	2,899	226	274	**2,850**	32,356
22	2048	5,246	5,612	2,942	224	280	**2,886**	35,242
23	2049	5,219	5,584	2,986	223	285	**2,924**	38,166
24	2050	5,193	5,556	3,030	222	291	**2,961**	41,127
25	2051	5,167	5,528	3,075	221	297	**2,999**	44,127
25 years		**137,283**	**146,871**	**64,744**	**10,192**	**12,350**		**44,127**

How each column is calculated

1. **Self-used, export** = PVsyst year-1 value × (1 − 0.5 %)$^{n-1}$ (5,828 / 6,235 kWh)
2. **Savings** = self-used kWh × 0.37 € × 1.02^{n-1}
3. **Export** = exported kWh × 0.0763 € (years 1–20), × 0.04 € after
4. **Costs** = 1 % × 18,462 € × 1.02^{n-1}, plus 2,500 € inverter (yr 13) and 9.84 × 400 € battery (yr 15)
5. **Net** = savings + export − costs; year 0 = −18,462 € investment

How the results follow

Payback = 7 + 768 ÷ 2,639 (cumulative crosses 0 in year 8)	**7.3 yr**
NPV = Σ Net$_n$ ÷ 1.03^n	**24,752 €**
IRR — discount rate at which NPV = 0	**12.7 %**
LCOE = Σ discounted costs ÷ Σ discounted kWh	**13.4 ct**

payback year replacement year

The case without battery uses the same model with 3,511 / 9,048 kWh, 13,050 € CAPEX and no battery replacement. Both sheets are in the Excel calculator.

All values from **PV_Residential_Economics_9000.xlsx**; every blue input cell can be changed and the model recalculates.

The battery stays worth it in every tested case

One input changed at a time

Case	Payback w/o	Payback with	NPV w/o	NPV with	Battery adds (NPV)
Base case	6.9	7.3	19,651	24,752	+5,101
Grid price 32 ct	7.6	8.3	16,061	18,794	+2,733
Grid price 42 ct	6.3	6.5	23,240	30,710	+7,470
Battery 400 €/kWh	6.9	6.7	19,651	26,547	+6,896
Battery 700 €/kWh	6.9	7.9	19,651	22,957	+3,306
PV 1,000 €/kWp	5.7	6.4	22,296	27,398	+5,102
PV 1,400 €/kWp	8.1	8.2	17,005	22,106	+5,101
Feed-in 5.2 ct	7.7	7.7	16,515	22,591	+6,076

Payback in years, NPV in € at 3 % over 25 years. Each row re-runs the full Excel model with one input changed.

Grid price drives everything

±5 ct moves the battery's added value between 2,700 € and 7,500 €.

Lower feed-in favours storage

At 5.2 ct (2027 draft) PV-only loses 3,140 €, the battery case only 2,160 €.

PV cost doesn't change the choice

It shifts both cases equally; the battery's added value stays 5,100 €.

RECOMMENDATION

Install PV with the 9.84 kWh battery

Highest NPV	24,752 €
Grid purchase	−42 %
Net bill, year 1	3,330 → 698 €
Payback	7.3 years

PV alone pays back 0.4 years faster, but the battery is a positive investment on its own (IRR 9.9 %, above the 3 % discount rate) and protects against falling feed-in tariffs. The tower takes up to 5 modules (16.4 kWh) if a 4th proves worthwhile.

What a real project would verify next

Topic	Current state in this study	Next step	Impact
Consumption	9,000 kWh = reference profile × 2, same shape all year	Smart-meter data (15-min); a heat pump or EV changes the shape	high
Roof size	House depth set to 9.40 m (real 8.14 m) so 25 modules fit	On the real roof: landscape layout, or use the north face / a garage	high
Battery ageing	Generic PVsyst model: 92.1 % health after year 1	Enter SMA cycle data (8,000 cycles) and re-run	medium
Prices	Market averages for 2026, not a quote	Two installer quotes; check KfW and local subsidies	medium
Battery size	3 modules; summer nights still need ~3.4 kWh from the grid	PVsyst variant with 4 modules (13.1 kWh) + Excel run	medium
Input A current	Imp 13.4 A above the 12.5 A of input A; PVsyst: 0.07 % loss	Confirm with SMA Sunny Design	low
Electrical details	No cable sizing or protection design yet	DC/AC cable cross-sections, SPD, meter cabinet	low

Deliverables of this study

Hand calculation (pages 04–08) · AutoLISP 3D scene generator · PVsyst reports for both cases · Excel calculator PV_Residential_Economics_9000.xlsx with 25-year cash flows and sensitivity

Skills shown

Load analysis · array, string, inverter and battery sizing · grid-code constraints · 3D modelling for shading · PVsyst simulation and loss analysis · self-consumption and storage analysis · investment appraisal (payback, NPV, IRR, LCOE)

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